

## SIMATS ENGINEERING ASSESSMENT TOOL 3

**COURSE NAME:** Database Management Systems  
**COURSE CODE:** CSA05

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### COURSE OUTCOME COVERED

**CO1:** *Analyze DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships. (BL1, BL2)*

Activity Tool : Technical Presentation (Low)  
Assessment Tool Weightage: 35%

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Dr

| QUESTIONS |                                                                                                                   |                  | Total Marks: 50 |
|-----------|-------------------------------------------------------------------------------------------------------------------|------------------|-----------------|
| Q.No      | Question                                                                                                          | Bloom's Level    | Marks           |
| 1         | Prepare a 5-slide presentation introducing DBMS: definition, objectives, advantages, and real-world applications. | BL2 – Understand | 5               |

|   |                                                                                                                        |                  |   |
|---|------------------------------------------------------------------------------------------------------------------------|------------------|---|
| 2 | Present a comparison between file-based systems and DBMS using real-life case studies from banking or healthcare.      | BL2 – Understand | 5 |
| 3 | Deliver a presentation explaining the three-level ANSI/SPARC architecture using diagrams and practical examples.       | BL2 – Understand | 5 |
| 4 | Present the different data models (Hierarchical, Network, Relational) with diagrams and industry use cases.            | BL1 – Remember   | 5 |
| 5 | Create and present an ER diagram for an Online Library System, explaining each component clearly.                      | BL2 – Understand | 5 |
| 6 | Prepare a presentation on the EER model, demonstrating generalization and specialization with a University case study. | BL2 – Understand | 5 |
| 7 | Present the concept of data independence (logical and physical) and its significance in DBMS architecture.             | BL2 – Understand | 5 |



|    |                                                                                                                                              |                  |   |
|----|----------------------------------------------------------------------------------------------------------------------------------------------|------------------|---|
| 8  | Deliver a technical presentation on the software components of a DBMS, explaining query processor, storage manager, and transaction manager. | BL1 – Remember   | 5 |
| 9  | Prepare a presentation comparing strong and weak entities in ER modeling with real-world examples.                                           | BL2 – Understand | 5 |
| 10 | Present a complete design walkthrough of an EER diagram for a Retail Chain Management System.                                                | BL2 – Understand | 5 |

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**Code of Conduct:**

I R Harika (192572140) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

R. Harika

**Signature of the Student.**

**RUBRICS FOR CALCULATION:**

| Technical Presentation     |           |                                                                    |                                                                |                                                          |                                                                     |
|----------------------------|-----------|--------------------------------------------------------------------|----------------------------------------------------------------|----------------------------------------------------------|---------------------------------------------------------------------|
| Criteria                   | Weightage | Level 4 – Exemplary                                                | Level 3 – Proficient                                           | Level 2 – Developing                                     | Level 1 – Beginning                                                 |
| Technical Content Accuracy | 30        | Highly accurate, in-depth, and insightful technical explanation    | Technically correct content with adequate depth and clarity    | Basic concepts presented with limited accuracy and depth | Content contains major technical errors; concepts poorly understood |
| Problem Identification     | 25        | Problem rigorously analyzed using appropriate and advanced methods | Problem clearly defined with a logical engineering approach    | Problem identified but analysis or methodology is weak   | Problem statement unclear or incorrect; no logical approach         |
| Use of Tools               | 20        | Advanced tools, well-analyzed data, and highly effective visuals   | Appropriate tools and data used effectively with clear visuals | Limited use of tools and basic data interpretation       | Minimal or incorrect use of tools and data; poor visuals            |



**SIMATS**  
**ENGINEERING**



**SIMATS**  
Saveetha Institute of Medical And Technical Sciences  
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

**CSA0501-DATABASE MANAGEMENT SYSTEM**

**CO3-TECHNICAL PRESENTATION**

**PRESENTED BY:**

R .HARIKA(192572140)  
(CSE-AI DEP)

**GUIDED BY:**

DR .A.JAYA MABEL RANI





# INTRODUCTION



1. Data is one of the most valuable resources in any organization.
2. A DBMS helps manage both small and large amounts of data.
3. It reduces manual work and improves efficiency.
4. DBMS supports decision-making by providing accurate information. Almost every organization today uses a database system.
5. Examples include schools, banks, hospitals, shopping websites, and government offices.



# DEFINATION



- A database is an organized collection of related data.
- DBMS acts as a bridge between users and the database.
- Users can insert, update, delete, and retrieve data easily.
- It supports multiple users at the same time.
- DBMS maintains relationships among different tables.
- It uses SQL (Structured Query Language) for database operations.



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# OBJECTIVE



1. Eliminate unnecessary duplication of data.
2. Ensure data is always accurate and up to date.
3. Provide quick access to required information.
4. Protect data from unauthorized users.
5. Maintain privacy of sensitive information.
6. Allow multiple users to work simultaneously.
7. Support backup and recovery after system failure.



# ADVANTAGES OF DBMS



1. Saves time in searching and updating records.
2. Supports data sharing among departments.
3. Prevents data inconsistency.
4. Provides centralized control over data.
5. Improves data quality.
6. Reduces maintenance cost.
7. Allows automatic report generation.
8. Supports cloud databases for remote access.



# Real-World Applications of DBMS



- **Banking:** Manages accounts, transactions, and customer information.
- **Education:** Stores student details, marks, and course information.
- **Healthcare:** Maintains patient history and medical reports.
- **Transportation:** Handles ticket booking and schedules.
- **Social Media:** Stores user profiles, posts, and messages.
- **Business:** Manages sales, inventory, and employee records.



## Topic 2: File-Based System vs DBMS



- File-Based System Example (Banking)
- Separate files may exist for customers, accounts, and transactions.
- Updating data in multiple files is time-consuming.
- Data inconsistency may occur due to duplicate data.
- Security is limited because files can be accessed easily.
- Data sharing between departments is difficult.





# Healthcare Case Study



- DBMS helps doctors quickly access patient information.
- Reduces paperwork and manual errors.
- Helps manage laboratory reports and prescriptions.
- Enables hospitals to track patient appointments.
- Improves healthcare service quality.

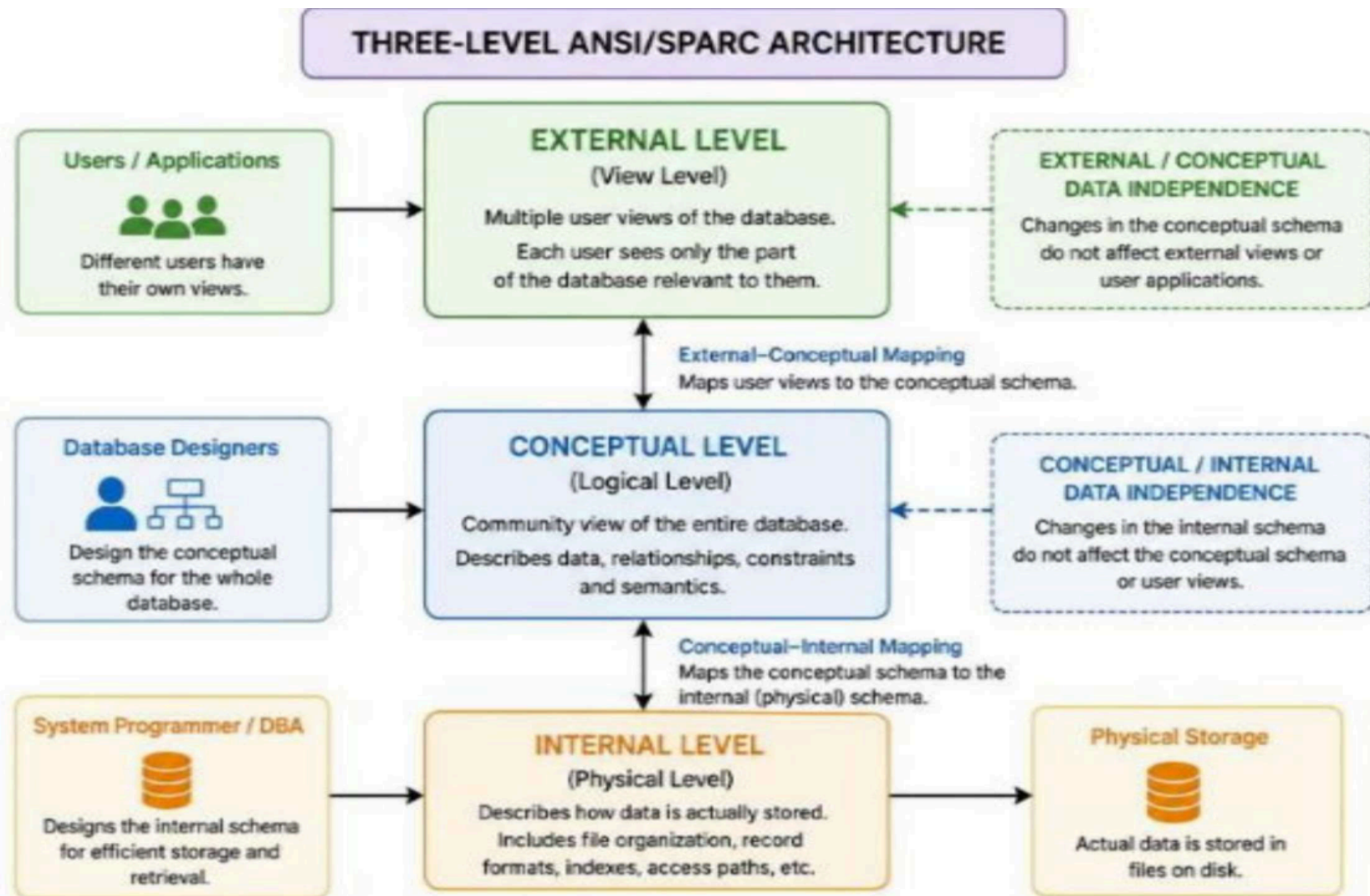




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### 3) ANSI/SPARC ARCHITECTURE



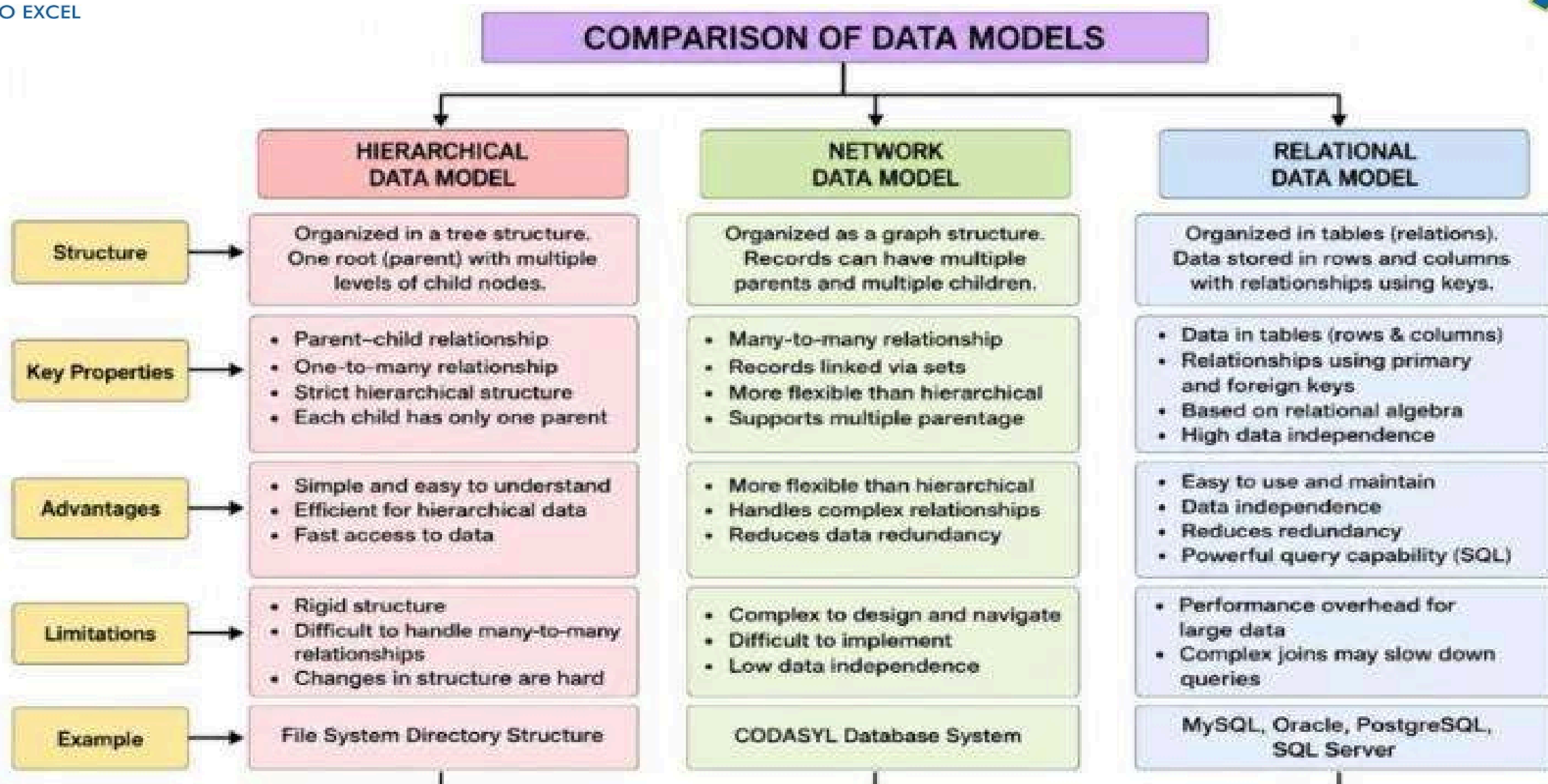


# EXAMPLE



- Student ID
- Student Name
- Department
- Marks
- Attendance
- Fee Details
- Faculty Details

## 4) COMPARISON OF DATA MODE



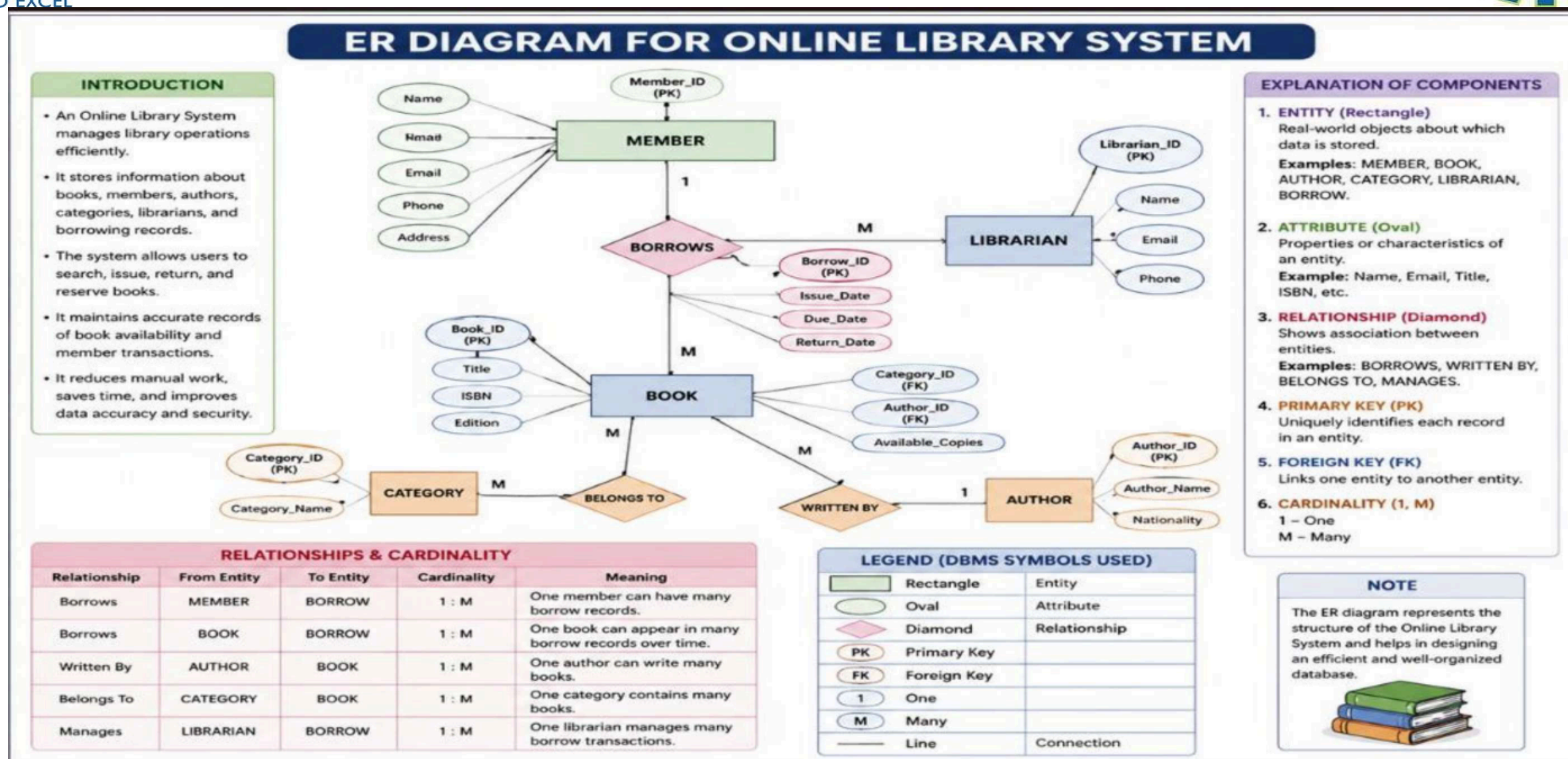




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# 5)ER DAIGRAM FOR ONLINE LIBRARY SYSTEM



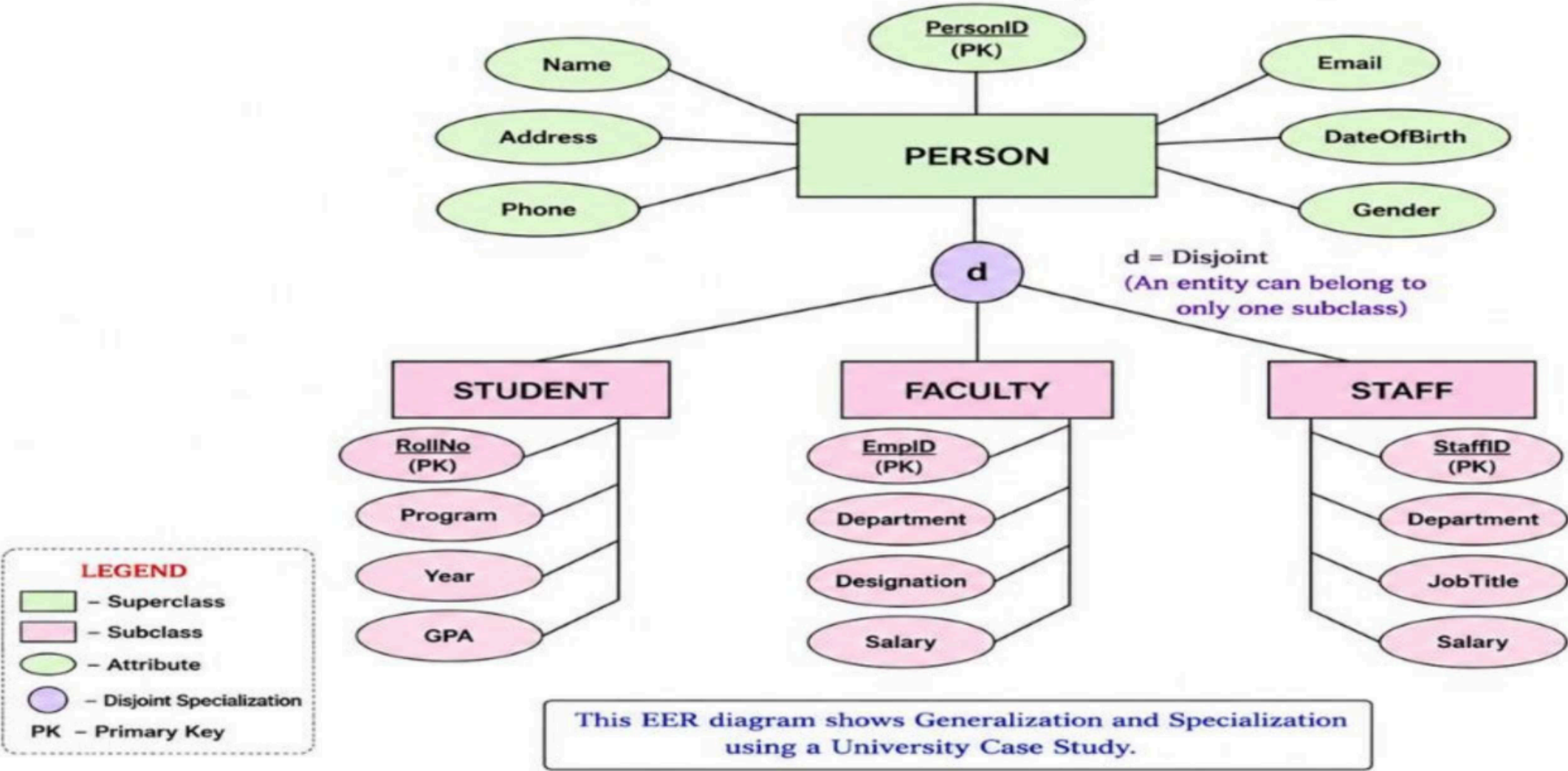


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# 6)EER DIAGRAM



EER DIAGRAM FOR UNIVERSITY MANAGEMENT SYSTEM  
(Generalization and Specialization)





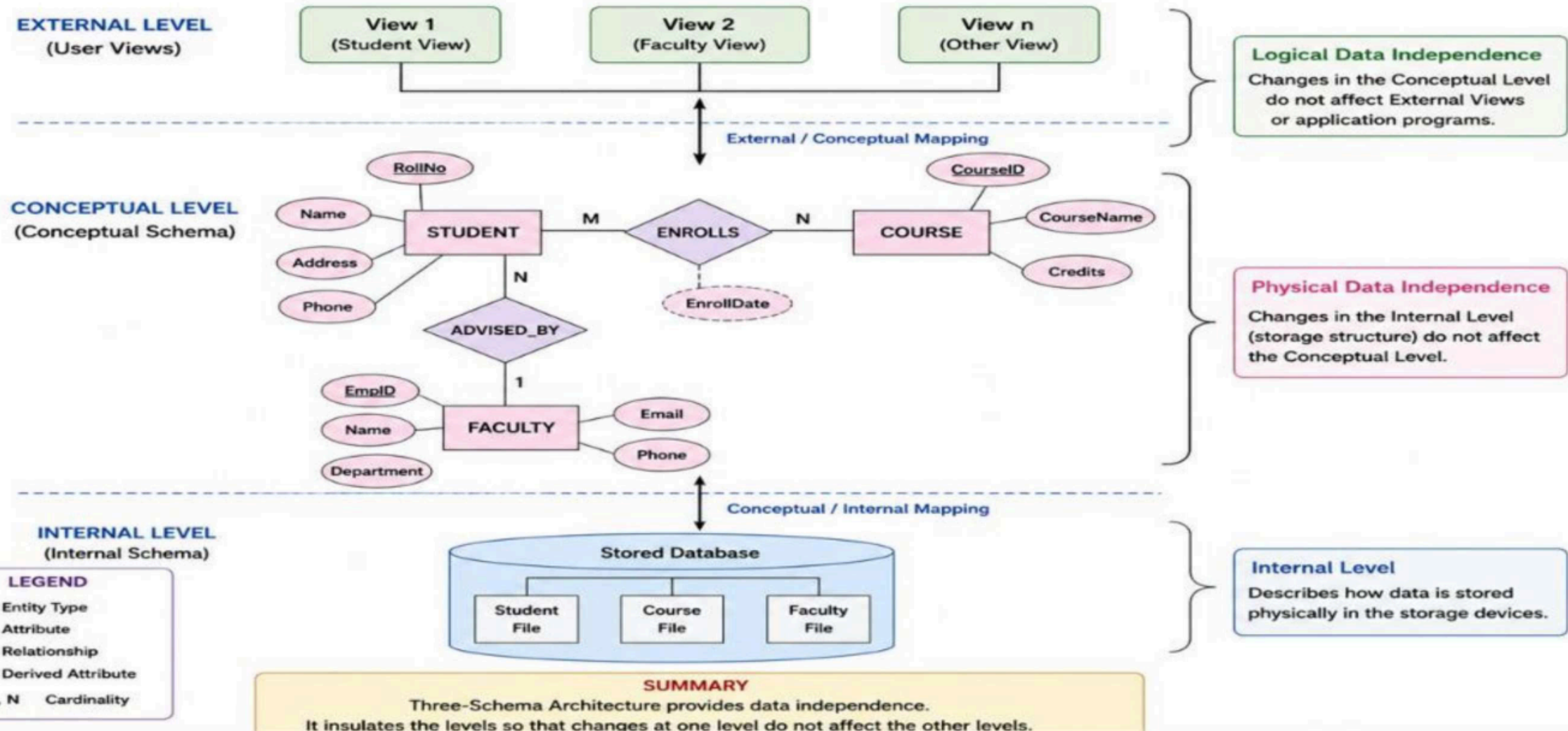


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# 7) DATA INDEPENDENCE IN DBMS

## DATA INDEPENDENCE IN DBMS – THREE-SCHEMA ARCHITECTURE



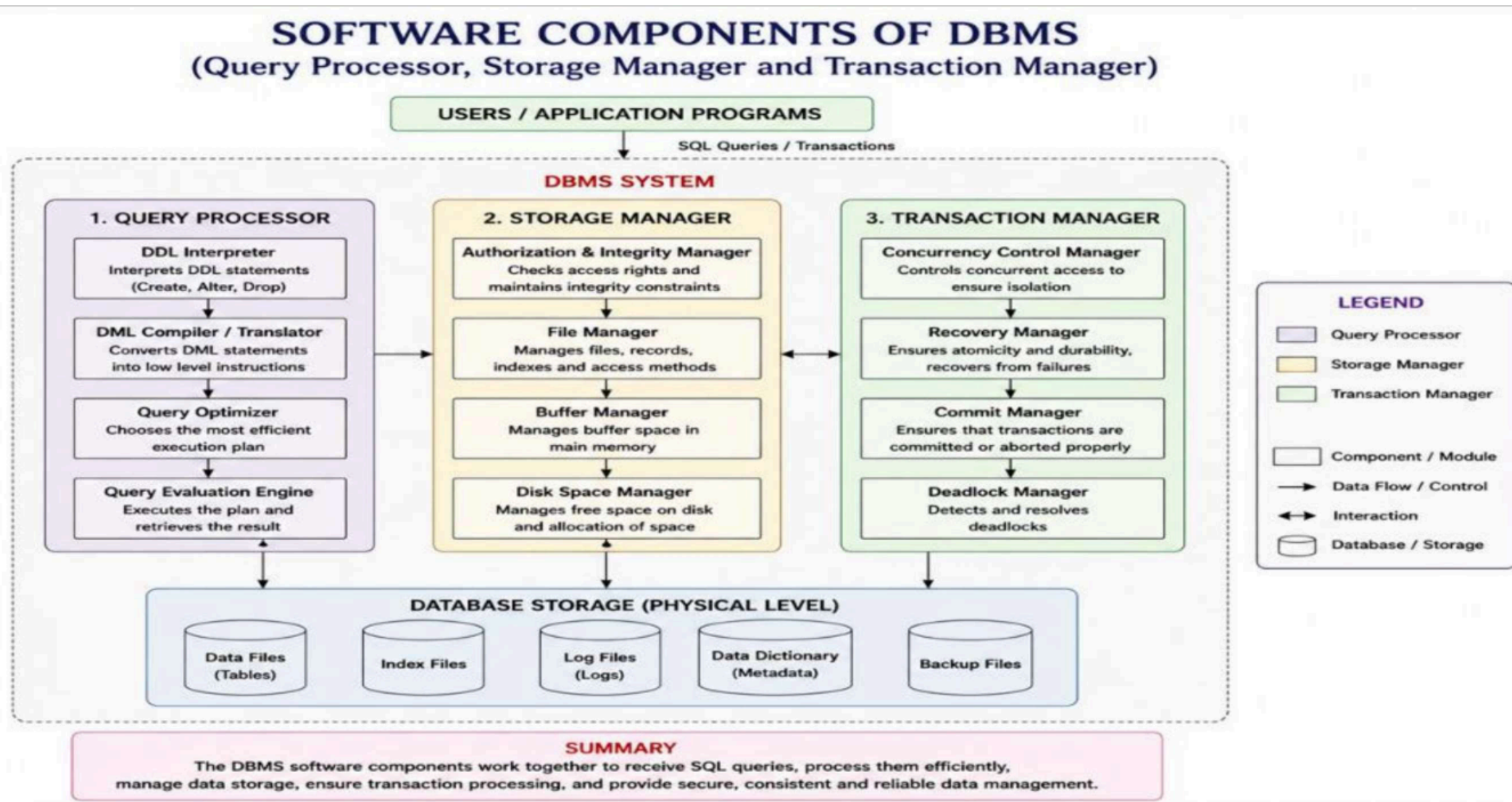




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# 8) SOFTWARE COMPONENTS OF DBMS





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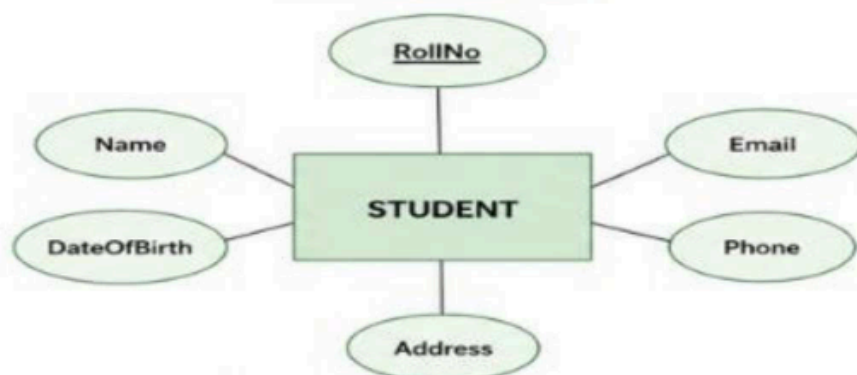
# 9)COMPARISON STRONG AND WEAK ENTITY

## COMPARISON OF STRONG AND WEAK ENTITIES IN ER MODELING

### STRONG ENTITY

- Has its own Primary Key.
- Can exist independently.
- Not existence-dependent on any other entity.

Example: STUDENT



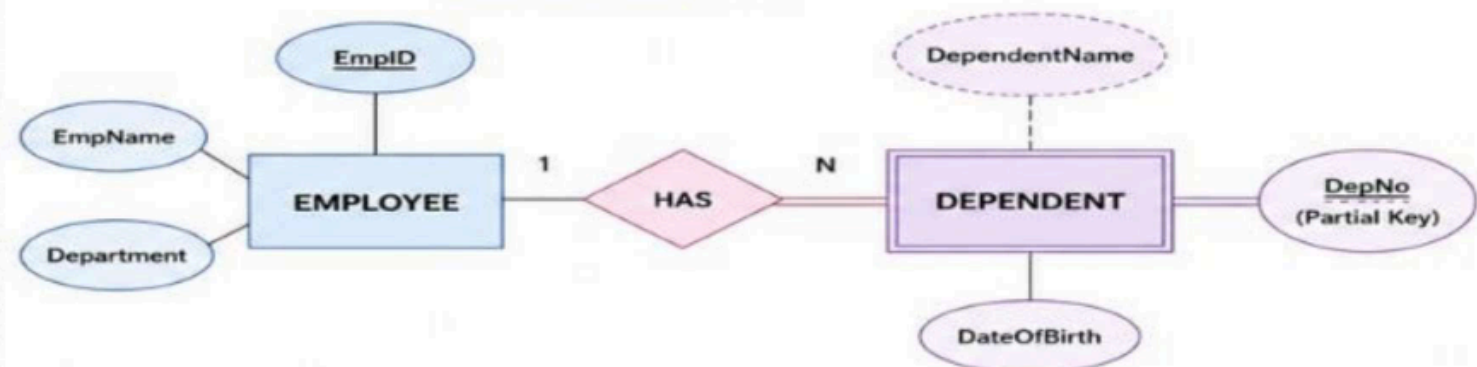
#### Explanation

STUDENT is a strong entity because it has its own primary key (RollNo) and can exist independently.

### WEAK ENTITY

- Does not have a complete Primary Key.
- Depends on a Strong Entity for its existence.
- Has a Partial Key (Discriminator).
- Connected to Strong Entity through Identifying Relationship.

Example: DEPENDENT



#### Explanation

DEPENDENT is a weak entity because it does not have its own primary key. It is identified using partial key (DepNo) along with EmpID of EMPLOYEE (Strong Entity) through the identifying relationship HAS.

### LEGEND



### REAL-WORLD EXAMPLES

- | Strong Entities | Weak Entities             |
|-----------------|---------------------------|
| • Student       | • Dependent (of Employee) |
| • Employee      | • OrderItem (of Order)    |
| • Department    | • Address (of Employee)   |
| • Customer      | • Payment (of Invoice)    |
| • Product       |                           |

Strong entities have independent existence, while weak entities depend on strong entities for their identification and existence.

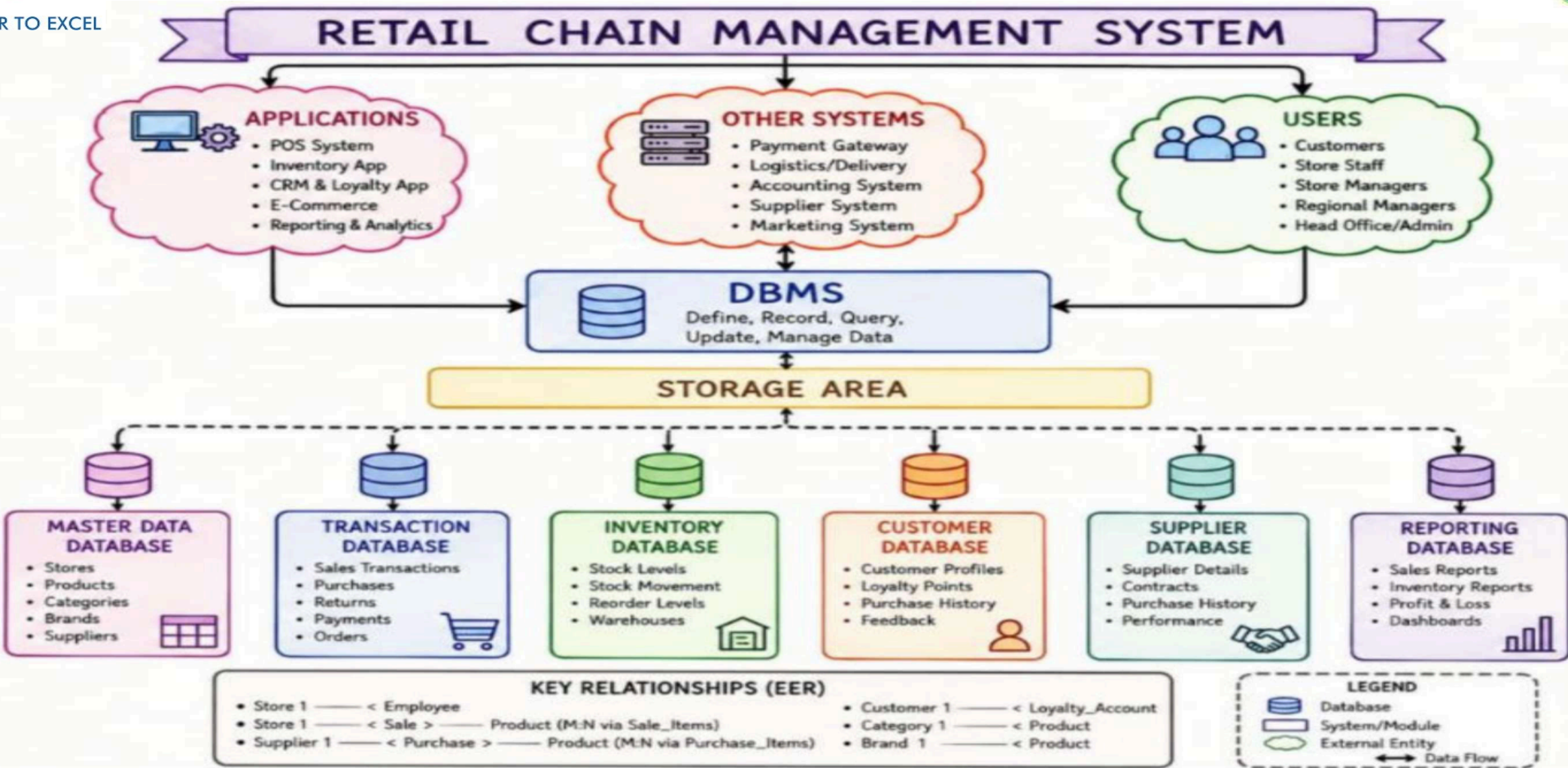




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# 10) RETAIL CHAIN MANAGEMENT SYSTEM





THANK YOU